

CIPHERMAIL: SECURE EMAIL SYSTEM WITH AUTONOMOUS SECOPS

PHASE I REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled “**CIPHERMAIL: SECURE EMAIL SYSTEM WITH AUTONOMOUS SECOPS**” is the bonafide work of **HEMNATH V (230701115), HARIPRASATH R S (230701103)**, who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

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ABSTRACT

communication is widely used for exchanging personal, academic, business, and confidential information. However, conventional email systems primarily provide encryption during transmission and storage, while the email provider may still technically access message content. Users also face security threats such as phishing links, sender spoofing, and dangerous email attachments.

This project proposes **CipherMail**, a Chrome browser extension designed to provide an additional privacy and security layer for Gmail. The system combines **end-to-end PGP email encryption** with an integrated **email threat-detection mechanism**. The encryption component uses OpenPGP.js to generate cryptographic keys and perform encryption, decryption, signing, and verification directly within the user's browser. A hybrid encryption approach uses a randomly generated AES session key for message encryption and the recipient's public key to protect the session key.

The threat-detection component analyzes email links, sender information, and attachment filenames using local heuristic rules and the VirusTotal API. The results are combined into Safe, Suspicious, or Dangerous classifications. CipherMail follows a no-backend architecture in which cryptographic operations are performed inside the browser and user data is stored locally using the Chrome storage API.

CONCLUSION

CipherMail demonstrates a combined privacy and threat-detection approach for Gmail using a Chrome Manifest V3 extension. The project implements PGP-based key management, encryption, decryption, signing and verification, together with a Gmail-integrated threat-scan action.

The privacy workflow is implemented locally using OpenPGP.js. The project does not implement cryptographic mathematics itself; instead, it uses a dedicated cryptographic library wrapper. Private keys remain encrypted at rest and are decrypted only briefly in memory when an operation requires them.

The threat-detection workflow provides local heuristic analysis of URLs, sender information and attachment names, supplemented by VirusTotal reputation checks for links. Results are combined into Safe, Suspicious or Dangerous classifications, and the user receives a warning before flagged actions.

The no-backend architecture is a significant design characteristic of the project. There is no CipherMail server or project-specific database. Browser-local storage is used for identities, contacts, settings, API keys and cached reputation results. The main external data transfer is the link URL sent to VirusTotal for reputation analysis.

The project also establishes clear boundaries. It does not claim protection against a compromised device, automatic interception of Gmail's native attachment upload flow, perfect phishing detection or a true hard block of Gmail's native actions. These limitations are explicitly considered part of the prototype's security and deployment model.